

CJAC_METHOD_V1 -- configuration ablation: pre-registration

Phase EXPERIMENT item 6 of the 2026-09-05 LOCK directive. DESIGN ONLY -- no execution until Andy's go on the live-call budget below. Copyright 2026 Andrew M Cohen. Apache 2.0.

Status: PRE-REGISTERED 2026-09-05. Any change to arms, metrics, ground truth, or analysis after execution begins is recorded as an amendment with its date and reason, never silently.

1. Question

Which stages of the corroboration pipeline earn their cost? Specifically: how much of the pipeline's ability to surface real defects in a rules node comes from (i) multiple model families, (ii) a separate adversarial pass, (iii) grounding (giving models only the cited source text) -- and what does each configuration cost in API dollars and in human triage minutes per node?

2. Ground truth (fixed before any arm runs)

- **Defect ledger:** rules/debt/validation/stage_b_dispositions.json at the debt-demo-v1.0 tag -- 120 node-specific dispositions plus 4 cross-cutting. Each carries classification and dangerous_direction. A ledger entry is a **known real defect** if its classification begins with FIXED (VERIFIED or SOURCE-NAMED) or is GLOSS-FOR-COUNSEL; it is a **known non-defect** if NOT-A-GAP or COVERED; HORIZON entries are excluded from both sets (real but out of scope).
- **Test articles (defect-bearing corpus snapshots), SHA-preserved in git:**
- **T38** = the 19 demo nodes at commit 27d5d44 (round 37, 2026-09-02) -- the content that carried the 58 round-38 findings (ledger entries with source "round 38" / run_20260903T174510Z).
- **T46** = the 19 demo nodes at commit b3d95db (round 45, 2026-09-05) -- the content that carried the 46 round-46 findings (run_20260904T221748Z). Each test article's defect set is the ledger entries whose source run was performed on that content. T38: 58 entries (15 nodes). T46: 46 entries (16 nodes). Materiality weight: DD = yes -> 2, otherwise 1.
- **Calibration fixtures:** the 14 replay fixtures (CAL-01..14) are used to confirm each arm's code path parses and scores before any live call; they are not part of the catch-rate denominator.
- **Fixed-bug history:** DAILY_CHANGELOG.md rounds 9-45 -- used only for the cost-of-alignment estimate (section 5), not for scoring.
- **Circularity, acknowledged and handled (self-review amendment, 2026-09-05, before execution).** The ledger's known defects were themselves found by arm (d) on T38 and T46, so arm (d)'s catch rate against the ledger alone is 100% by construction and the other arms would be scored against (d)'s own output. Therefore the scoring denominator is NOT the ledger as it stands. It is the **UNION** of (i) the ledger's known real defects for that article and (ii) every material finding any arm produces on that article that Cowork and Andy classify as a real defect after review (same classification discipline as rounds 38-46). Catch rate for every arm, including (d), is measured against that union; findings that only (d) produced are still credited to (d), and findings that (d) missed but (b) found count against (d). The pre-run ledger-only figure is ALSO reported, labelled "ledger-only (favours (d))", so the reader can see both. This changes the execution order: all arms run on an article, THEN classification, THEN scoring -- no arm is scored before the union is closed.

3. Arms

All arms run on the SAME test articles with the SAME adversarial output schema (the round-44 three-flag schema; see 3.1) so findings are comparable. Model identifiers frozen at execution time and recorded.

Arm	Derivation	Verification / adversarial	Grounding
(a) single + self-critique	claude-opus-5 grounded derivation	same model, structured self-critique prompt (the Stage B prompt addressed to its own derivation)	enforced
(b) single + cross-family adversarial	claude-opus-5 grounded derivation	gemini-2.5-pro runs the Stage B prompt	enforced
(c) two-model draft/verify	claude-opus-5 grounded derivation	gpt-5.5 grounded re-derivation + LLM judge + gpt-5.5 Stage B	enforced
(d) tri-model consensus (current pipeline)	3 families grounded derivation + judge	claude-opus-5 Stage B	enforced
(e) ungrounded control	winner of (a)-(d), same models, source text WITHHELD (models answer from training knowledge given only the node's cite labels)	as winner	none

3.1 Finding-format variable (held round 47). The pipeline currently asks for three yes/no flags per edge case and every node returns three material findings. Arm (d) is additionally run once with the round-47 schema (severity rank 1-5 + "must-fix for a careful legal-aid attorney" flag) on T46 only, to measure whether the format changes catch rate or false-flag rate. This is the only place round 47 enters before v1.1.

4. Metrics (per arm, per test article)

1. **Catch rate** = sum of weights of real defects (the UNION set of section 2) the arm reports / sum of weights of all real defects in the union. A defect is "reported" when a finding substantially overlaps the ledger entry's theme -- decided by an LLM matcher (a model family NOT used in the arm) given the finding and the ledger entry, with Cowork reviewing every match and every near-miss and Andy spot-checking 20% of matches. Matcher decisions are archived.
2. **False-flag rate** = findings that match a NOT-A-GAP/COVERED ledger entry, or that Cowork+Andy classify as non-material on review, / total findings. This is the alignment-cycle cost driver.
3. **New-defect yield** = material findings that match no ledger entry (reported, classified, and sent to POST_V1_BACKLOG.md -- v1.0 content untouched).
4. **API cost per node** from _usage tokens x published per-token prices at execution date, plus citation-check cost (zero) -- recorded per stage.
5. **Projected human minutes per node** = findings x observed triage minutes per finding (rounds 38-46: Cowork ~12 min/finding to source-pin and disposition; Andy's audit ~40 min/node from the census estimate) + fixed review overhead.
6. **Stage A agreement and citation verification** reported alongside for arms that have them.

Dual-reported: raw and after matcher review, per house rule.

5. Replay-first execution plan and budget

- • **Zero-cost first:** arm (d) on T38 and T46 already exists as run_20260903T174510Z and run_20260904T221748Z. Their findings are matched to the ledger offline -> arm (d) catch and false-flag rates without a single new call. The matcher itself is the first live spend (~\$3).
- • **Snapshot loading:** runner gains --corpus-ref <git-ref> (reads node JSON from git show <ref>:<path>) and --config a|b|c|d and --ungrounded; all replay-tested against the 14 fixtures before live use (calibration must be green -- constraint).
- • **Smoke:** each new arm on 3 nodes of T46 first (~\$1 per arm).

• **Live budget proposal (cap):** arms (a), (b), (c), (e) x 2 test articles x 19 nodes at \$0.20-0.35/node $\approx$ \$45; arm (d) round-47 variant on T46 $\approx$ \$9; matcher and judge calls $\approx$ \$6. **Proposed cap: \$70**, in four gated tranches (offline matcher \$3 -> smoke \$5 -> T46 arms \$30 -> T38 arms \$30), each needing Andy's go and run by Andy. Per-run estimates printed before every run as today.

6. Pre-registered predictions (so the result can disagree with us)

P1. Cross-family adversarial (b) catches materially more than self-critique (a) on the same model. P2. The tri-model consensus stage (d vs c) adds little to CATCH rate (Stage A has been 100% for weeks) but is what produces the citation-verified derivation record; its value is evidentiary, not defect-finding. P3. The ungrounded arm (e) produces MORE findings and a HIGHER false-flag rate (it hallucinates rules and then "finds" the node disagrees with them) and misses the citation-transcription defects entirely. P4. The severity-rank format lowers false-flag rate without lowering catch rate on DD defects. If P2 holds, the recommended scaling configuration is (b) or (c) for defect-finding with (d)'s derivation record generated once per release, not per iteration.

7. Analysis and reporting

Per arm and test article: the four rates with n, cost per node, projected minutes per node, and the DD-only catch rate. A single table, plus the cost curve: cumulative API dollars and human minutes per validated node under each configuration. Results reported regardless of outcome; if an arm's code path fails, the failure is the result for that arm and is reported as such.

Deliverable: docs/CJAC_METHOD_V1.md -- written for an external technical reader (a clinic, a lab, a replicating team): the pipeline as a recipe (inputs, stages, prompts by reference, gate definitions), the evidence table above ranking what each stage earns, the recommended configuration for scaling with its cost per validated node, the known limitations (one corpus, one jurisdiction pair, model versions of record), and the replication procedure (this repo, these commands, this budget).

8. Constraints restated

One-variable rule. No live runs by anyone but Andy; smoke first. v1.0 is immutable -- experiment findings about v1.0 content go to POST_V1_BACKLOG.md. Calibration green before any experiment code runs live. Budget caps and per-run estimates as above. This document is amended, never overwritten.